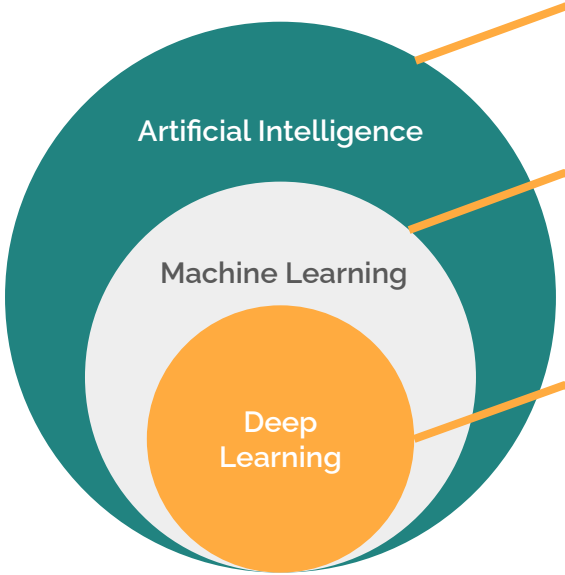
Muzafar Rasool Bhat
Sr. Assistant Professor
Department of Computer Science
Islamic University of Science and Technology



Machine Learning

Basics

Branches of Artificial Intelligence

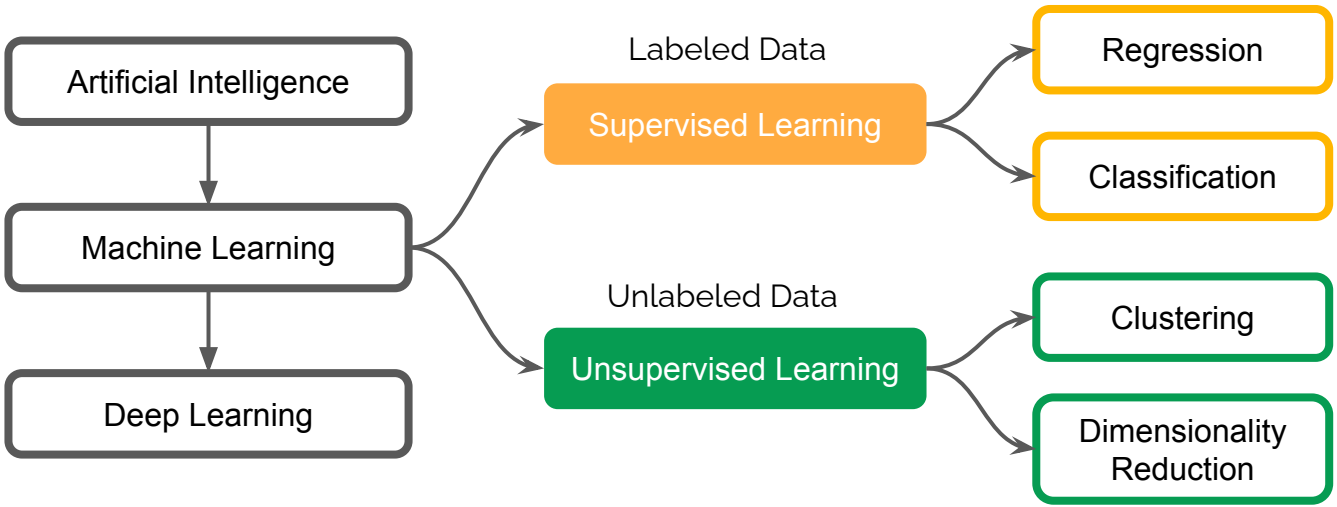


Artificial intelligence is the simulation of human intelligence processes by machines, especially computer systems.

Machine learning is a sub field of study in artificial intelligence in which machines learn problem solving without being explicitly programmed.

Deep learning is a field of study in machine learning that allows computational models that are composed of multiple processing layers to learn representations of data with multiple levels of abstraction.

Machine Learning Subdomains



Learning Paradigms

Supervised Machine Learning



Dog

Cat

Supervised Machine Learning Algorithm



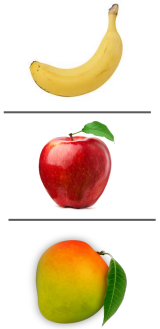
Supervised Machine Learning Algorithm

Cat

Unsupervised Machine Learning



Unsupervised Machine Learning Algorithm



Category 1

Category 2

Category 3



New Instance

Regression Analysis

Linear Function

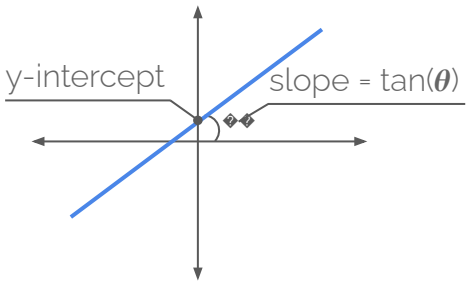
$$y = mx + c \quad \text{or}$$

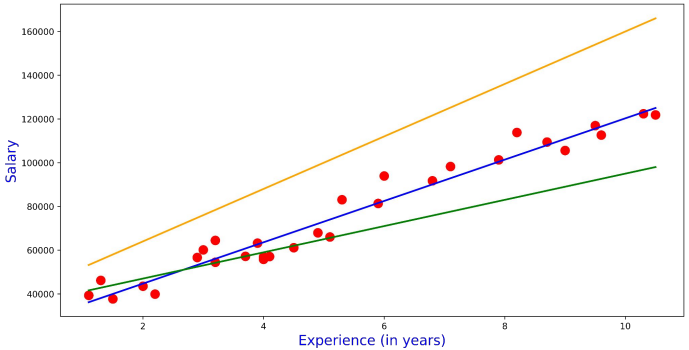
$$h_\theta(x) = \theta_0 + \theta_1 x$$

where,

$\theta_1 = m$ is slope

$\theta_0 = c$ is y-intercept





Model 1

`f(x) = 12000*x+40000`

Model 2

`f(x) = 6000*x+35000`

Model 3

`f(x) = 9449*x+25792`

Gradient Descent

$$Hypothesis : h_\theta(x) = \theta_0 + \theta_1 x$$

$$Parameters : \theta_0, \theta_1$$

$$CostFunction : J(\theta_0, \theta_1) = \frac{1}{2m} \sum_{i=1}^{m} (h_\theta(x^{(i)}) - y^{(i)})^2$$

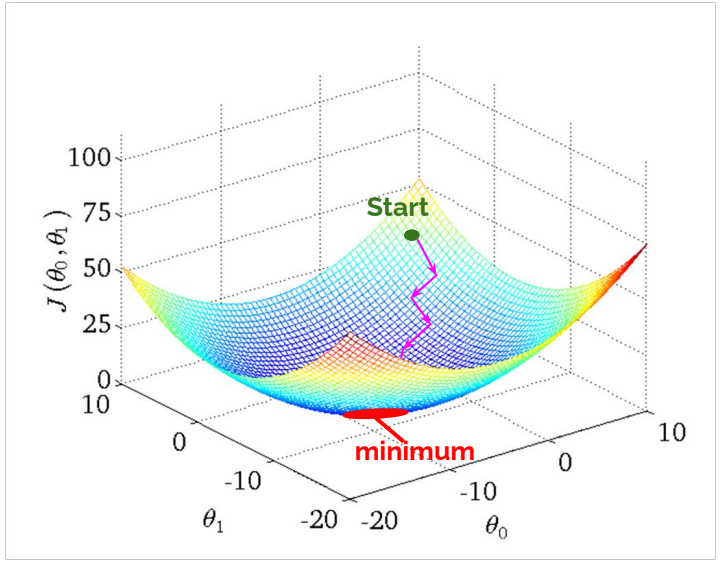
$$Goal : \min_{\theta_0, \theta_1} J(\theta_0, \theta_1)$$

Gradient Descent

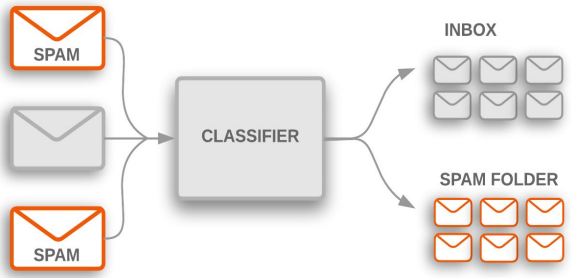
repeat until convergence {

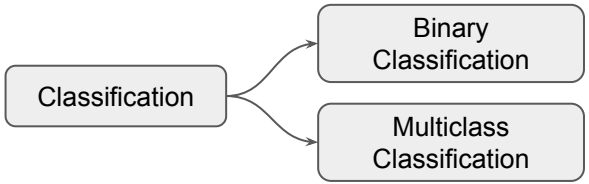
$$\theta_j := \theta_j - \alpha \frac{\partial}{\partial \theta_j} J(\theta_0, \theta_1)$$

}



Classification

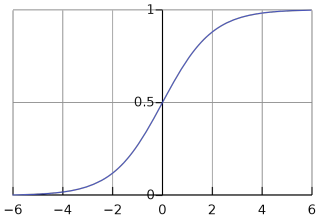




Logistic Regression Model

$$h_\theta(X) = \theta^T X$$ Linear Model

$$0 <= h_\theta(X) <= 1$$

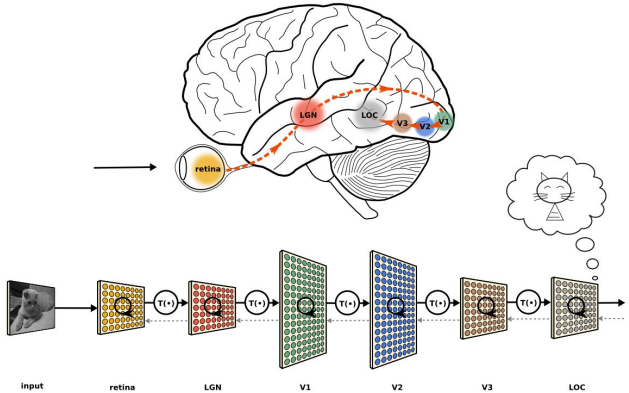


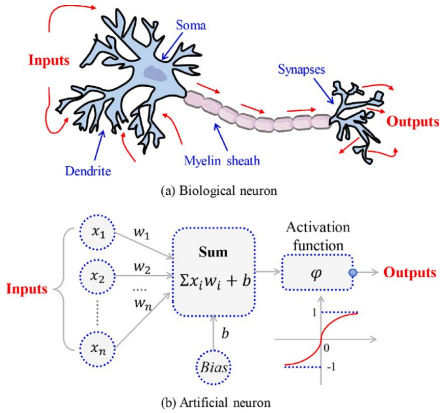
$$h_\theta(X) = g(\theta^T X)$$

$$where,\ g(z) = \frac{1}{1+e^{-z}}$$

Sigmoid Activation Function

Deep Learning





LGN (Lateral Geniculate Nucleus): acts as a relay center for visual information coming from the optic nerves and sends it to the primary visual cortex (V1).

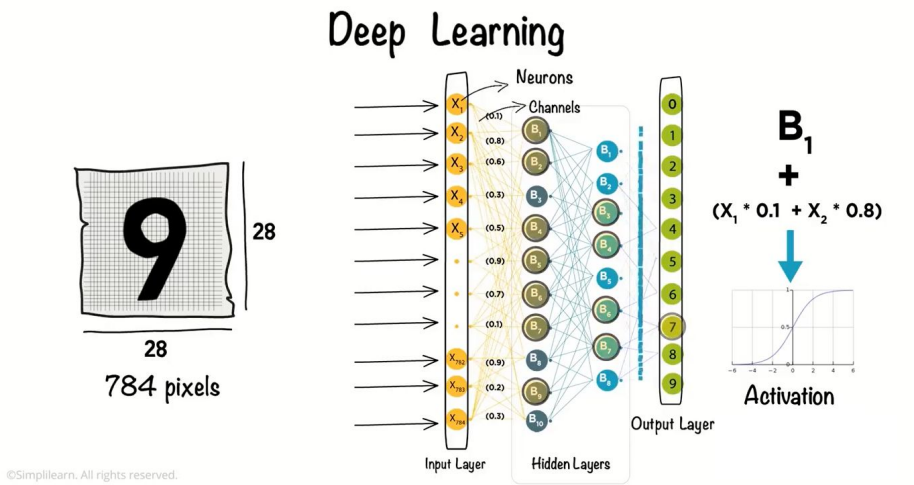
V1 (Primary Visual Cortex): It is the primary processing center for visual information and is responsible for basic visual processing, such as edge detection, orientation, and simple motion detection.

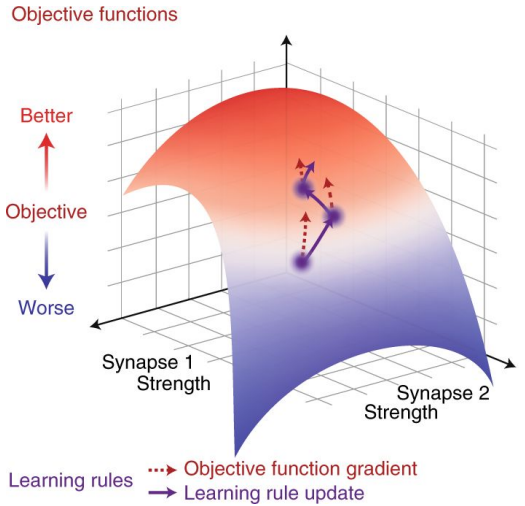
V2 (Secondary Visual Cortex): V2 is adjacent and secondary to V1. It further processes visual information after it leaves V1. V2 continues the processing of visual features such as orientation, spatial frequency, and color.

V3 (Visual Area 3): It is adjacent to V2 and contributes to more complex visual processing tasks beyond those performed in V1 and V2.

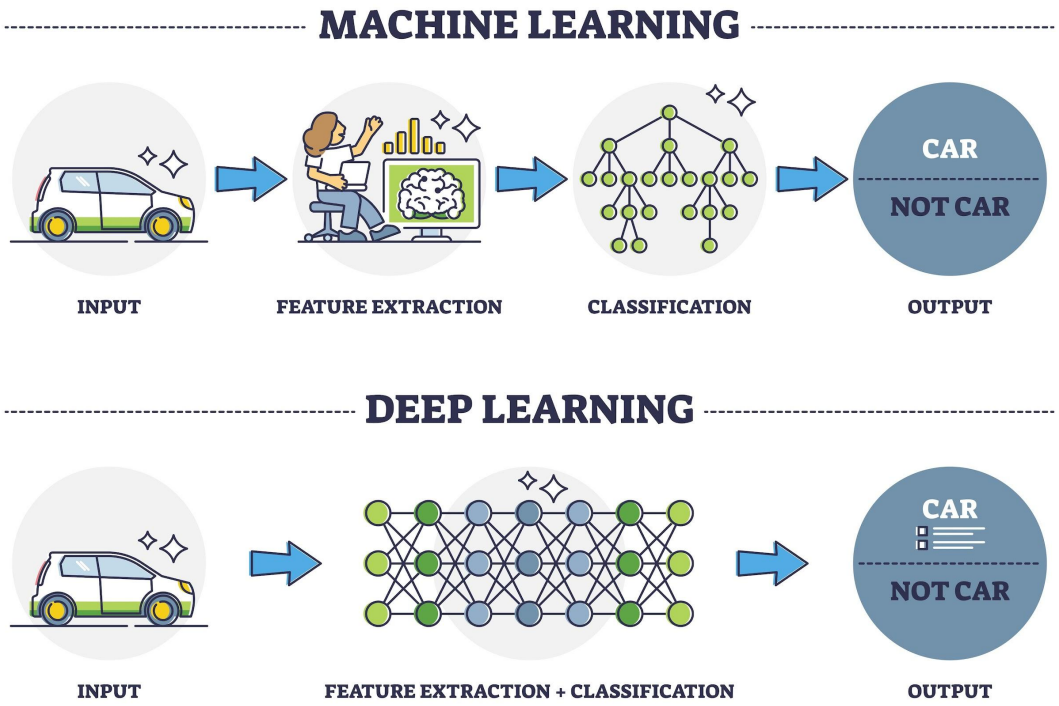
LOC (Lateral Occipital Complex): LOC is not strictly a part of the hierarchy of early visual processing regions like LGN, V1, V2, and V3. Instead, it is located in the lateral occipital lobe and is involved in object recognition and shape processing. It integrates visual information from different parts of the visual field and is crucial for recognizing objects and shapes.

Deep Learning Working





How Deep Learning is different from Machine Learning





Thanks

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